# P-ADIC INTEGERS, P-ADIC NUMBERS AND HASSE-MINKOWSKI THEOREM

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ABSTRACT. This paper gives an introduction to p-adic integers, discussing their algebraic structure in relation to groups, rings, and its field of fractions also known as p-adic numbers. It then states, but not prove, Hasse-Minkowski Theorem with connections to p-adic numbers and provide examples to showcase its usefulness in solving for homogeneous quadratic forms. Lastly, the paper briefly discusses further p-adic applications to geometry, called Monsky's Theorem, and to quantum mechanics.

## 1. Introduction & Motivation

P-adic integers and p-adic numbers were initially invented by German Mathematician Kurt Hensel in 1897 when he was studying algebraic number fields [1]. It has since developed significantly with applications to many fields such as number theory, geometry, cryptography and even quantum mechanics. To motivate its study, consider the seemingly simple homogeneous quadratic form $x^2 + y^2 - 3z^2 = 0$ where $x, y$ and $z$ are variables. Does this equation admit rational solutions over the field $\mathbb{R}^3$? If not, how would we go about proving it? With the help of p-adic integers and p-adic numbers, Hasse formulated the Hasse-Minkowski Theorem that answered this question. We will begin with the definition and algebraic structure of p-adic integers in Section 2, construct its field of fractions known as p-adic numbers in Section 3, explore the Hasse-Minkowski Theorem in Section 4, and discuss further applications of p-adic numbers in Section 5.

## 2. Ring $Z_p$ of P-adic Integers

The definitions and theorems stated here will closely follow a p-adic graduate textbook by A. M. Robert [2]. It is a great textbook that is also found free online. We start by recalling the definition of a formal series and a formal power series.

**Definition 1.** A formal series is an infinite sum that has no notion of convergence and is manipulated by the usual algebraic operations $(+, -, \div, \times)$ on a series.

**Definition 2.** Let $R$ be a ring. A formal power series is a special kind of formal series of the form

$$\sum_{i \geq 0} a_i x^i = a_0 + a_1 x + a_2 x^2 + \dots$$

where each $a_i \in R$.

With this, we can now give the definition of p-adic integers.

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**Definition 3.** Let p be a prime. A p-adic integer is a formal power series of the form
$$\sum_{i\geq 0} a_i p^i = a_0 + a_1 p + a_2 p^2 + \dots$$
where each $a_i \in \mathbb{Z}_p$.

Notation warning: Be careful to note we will represent $Z_p$ as the set of p-adic integers, while $\mathbb{Z}_p$ will represent the usual integers modulo p.

**Example 4.** The following is an element of 2-adic integers $Z_2$
$$1 \cdot 2^0 + 1 \cdot 2^2 + 1 \cdot 2^{80}$$
Note that all the coefficients are in $\mathbb{Z}_2$.

A point of interest for the set of p-adic integers is that they form groups and rings under the standard notion of addition and multiplication. But before that, we introduce the idea of 'system of carry'. Let p be a prime. For the addition of two coefficients $a, b \in \mathbb{Z}_p$, it is $a + b$ if it is less than or equal to $p - 1$, otherwise it is $a + b - p$ and we carry a one to the next power. The following example makes this idea clear.

**Example 5.** Let $a, b \in Z_p$ such that
$$a = 1 = 1 + 0p + 0p^2 + \dots$$
$$b = (p-1) + (p-1)p + (p-1)p^2 + \dots$$
Then observe that $a + b$ has the first term $1 + (p - 1) = p$, which gets carried over to the second term. Subsequently, this keeps getting carried over, causing all terms to be 0. Thus, we have that $a + b = 0$.

**Theorem 6.** *Let $Z_p$ be the set of p-adic integers. The set $Z_p$ with the standard binary operation $+ : R \times R \to R$ defined component-wise with the system of carry as*
$$\sum_{i\geq 0} a_i p^i + \sum_{i\geq 0} b_i p^i = \sum_{i\geq 0} (a_i + b_i) p^i$$
*is an abelian group.*

*Proof.* We will skip through closure and abelian as these are inherited from the standard addition operation with the additional idea of system of carry. For associativity, we can first perform the standard operations without the system of carry, finding them to be equal, which is then also equal after applying the system of carry. For additive identity, note we can take $0 = \sum_{i\geq 0} 0p^i \in Z_p$. For additive inverse, consider arbitrary $a = \sum_{i\geq 0} a_i p^i \in Z_p$. Take $-a = 1 + \sum_{i\geq 0}(p-1-a_i)p^i \in Z_p$ as the additive inverse. This construction is valid since if $a_i \in \mathbb{Z}_p$, then necessarily $p - 1 - a_i \in \mathbb{Z}_p$. We see that $a - a = \sum_{i\geq 0} a_i p^i + 1 + \sum_{i\geq 0}(p - 1 - a_i)p^i = 1 + \sum_{i\geq 0}(p-1)p^i = 0$, using the result from previous example. □

**Theorem 7.** *Let $Z_p$ be the set of p-adic integers. The set $Z_p$ with the standard binary operations $+ : R \times R \to R$ component-wise and $\times : R \times R \to R$ defined with the system of carry as*
$$\sum_{i\geq 0} a_i p^i + \sum_{i\geq 0} b_i p^i = \sum_{i\geq 0} (a_i + b_i) p^i$$

$$\sum_{i\geq 0} a_i p^i \times \sum_{i\geq 0} b_i p^i = \sum_{i\geq 0} \left( \sum_{j+k=i} a_j b_k \right) p^i$$

*is a ring.*

*Proof.* We have already proven it is an abelian group with an additive identity of $0 = \sum_{i\geq 0} 0p^i$. We note that it is commutative as it inherits the property from the standard multiplication operation. For multiplicative identity, take $1 = 1p^0 + 0p^2 + 0p^3 + \cdots \in Z_p$. For associativity and distributivity, we argue similarly as in previous theorem. We can again perform the operations without the system of carry, having both sides equal, then apply the system of carry, which is again equal. □

Notation: We will take $Z_p$ as the ring of p-adic integers with the defined binary operations from here on.

Let's look at the following example to ensure we understand the operations of $Z_p$.

**Example 8.** Consider the two elements $a, b \in Z_3$

$$a = 2 \cdot 3^0 + 1 \cdot 3^1 + 2 \cdot 3^2$$

$$b = 2 \cdot 3^1 + 1 \cdot 3^3$$

Then with the system of carry,

$$\begin{aligned} a + b &= 2 \cdot 3^0 + 3 \cdot 3^1 + 2 \cdot 3^2 + 1 \cdot 3^3 \\ &= 2 \cdot 3^0 + 0 \cdot 3^1 + 0 \cdot 3^2 + 2 \cdot 3^3 \\ &= 2 + 2 \cdot 3^3 \end{aligned}$$

$$\begin{aligned} ab &= 4 \cdot 3^1 + 2 \cdot 3^2 + 2 \cdot 3^3 + 4 \cdot 3^3 + 1 \cdot 3^4 + 2 \cdot 3^5 \\ &= 1 \cdot 3^1 + 0 \cdot 3^2 + 1 \cdot 3^3 + 0 \cdot 3^4 + 0 \cdot 3^5 + 1 \cdot 3^6 \\ &= 1 \cdot 3^1 + 1 \cdot 3^3 + 1 \cdot 3^6 \end{aligned}$$

## 3. Construction of Fraction Field $Q_P$

**Theorem 9.** *Let $p$ be a prime. The ring of p-adic integers $Z_p$ is a domain.*

*Proof.* First note that the ring $Z_p$ is never $\{0\}$. To show it is a domain, we need to show it has no non-zero zero divisors. Assume for contradiction that it does. Let $\alpha = \sum_{i\geq 0} a_i p^i \in Z_p$, $\alpha \neq 0$, has a non-zero zero divisor $\beta = \sum_{i\geq 0} b_i p^i \in Z_p$, such that $\alpha\beta = \sum_{i\geq 0} 0p^i$. Let $a_k$ and $b_m$ be the first non-zero coefficient of $\alpha$ and $\beta$ respectively. Since $p$ does not divide both $a_k$ and $b_m$, it does not divide their product $a_k b_m$ either. Therefore, we require $b_m = 0$. We can recursively use this argument now with $b_{m+1}$ and get that $\beta = 0$, a contradiction. □

Hence, we can construct the p-adic field as the field of fractions of $Z_p$, denoted $Q_p$. Note that $Q_p$ is also commonly known as the p-adic numbers and an element $a \in Q_p$ can be written in the form $\sum_{i\geq m} a_i p^i$ for some $m \in \mathbb{Z}$ and $a_i \in \mathbb{Z}_p$. For reasons as to why we can write it in this form, check out the textbook [2, pg. 36].

**Example 10.** The following are elements of $Q_7$:

$$2 \cdot 7^{-3} + 5 \cdot 7^{-1} + 3 \cdot 7^2 + 6 \cdot 7^6, \quad 6 \cdot 7^{-40} + 1 \cdot 7^4 + 4 \cdot 7^{23}$$

The following propositions describe natural embeddings where any rational number can be represented as an element in the p-adic field, subsequently leading to p-adic field having a nice property of characteristic 0.

**Proposition 11.** *Let p be a prime.*

(1) *There exists an injective ring homomorphism* $\mathbb{Z} \hookrightarrow Z_p$.
(2) *There exists an injective field homomorphism* $\mathbb{Q} \hookrightarrow Q_p$.
(3) $Q_p$ *is a field of characteristic 0.*

*Proof.* For simplicity sake, we will only outline the proof for each cases. For (1), we can essentially find an embedding $\mathbb{Z} \hookrightarrow Z_p$, see textbook [2, Ch.4] for the details. For (2), consider the injective map $\phi : \mathbb{Z} \hookrightarrow Z_p$ and define the map $\psi : \mathbb{Q} \to Q_p$ such that for any rational number $q = \frac{a}{b}, a, b \in \mathbb{Z}, b \neq 0$, we have $\psi(q) = \frac{\phi(a)}{\phi(b)} \in Q_p$. As you can check, this map $\psi$ is well-defined and is an injective field homomorphism. For (3), since $\mathbb{Q}$ has characteristic 0 and that $\mathbb{Q} \hookrightarrow Q_p$, $Q_p$ also has characteristic 0. □

## 4. P-adic Integers in Number Theory: Hasse-Minkowski Theorem

The p-adic field $Q_p$ has a big application to solving homogeneous quadratic Diophantine equations, or more broadly quadratic forms, in Number Theory through Hasse-Minwkoski Theorem. It helps breakdown problems in $\mathbb{Q}$ to problems in $\mathbb{R}$ and $Q_p$. This conversion of a global $\mathbb{Q}$ field into local $\mathbb{R}$ and $Q_p$ fields is how the theorem also got its name as the Local-Global Principle, where often times it is easier to study the local fields rather than the global field. It should be strongly noted that the Hasse-Minkowski Theorem only holds for homogeneous quadratic forms. We begin by recalling what Diophantine equations and quadratic forms are.

**Definition 12.** A Diophantine equation is a polynomial equation with integer coefficients for which only integer solutions are of interest.

**Definition 13.** A quadratic form in $n$ variables over a field K is a function f of the form

$$f(x_1, x_2, \ldots, x_n) = \sum_{i \leq j}^{n} a_{ij} x_i x_j$$

where $a_{ij} \in K$.

**Example 14.** Equations of the form $x^2 + 2xy + 2y^2 - z^2,\ x^2 + 4y^2$ over field $\mathbb{R}$ is of quadratic form.

**Theorem 15.** *(Hasse-Minkowski Theorem). Let f be a quadratic form in n variables over* $\mathbb{Q}$. *Then* $f = 0$ *has a non-trivial solution in* $\mathbb{Q}^n$ *if and only if* $f = 0$ *has a non-trivial solution over* $\mathbb{R}^n$ *and* $Q_p^n$ *for all prime p.*

*Proof.* The forward direction is trivial. Since $\mathbb{Q} \subset \mathbb{R}$ and $\mathbb{Q} \subset Q_P$ for all prime p through the embedding of proposition 11, finding a rational solution implies finding a solution in $Q_p$ and $\mathbb{R}$. For the reverse direction, we will omit the proof as it is too long. Look at Serre's book [3, p. 41–43] for the full proof. □

It is worth pointing out that because of homogeneity of the quadratic form, we can multiply the lowest common multiple of all the solution combinations of denominators to lift rational solutions into integer solutions, hence solving the Diophantine equation.

Of course, the significance of this theorem comes greatly in reducing the ease to checking whether a quadratic form admits rational solution. Note that when checking for solutions in $Q_p$, it suffices to consider solutions in $Z_p$. This is again due to the homogeneity of quadratic forms, where all non-trivial solutions in $Q_p$ can be scaled by a power of p such that all coefficients lie in $Z_p$ with at least one coefficient not divisible by p, that is the $p^0$ term. A trivial solution to Hasse-Minkowski over $Q_p$ would then have the condition $a \in Q_p, a \equiv 0 \pmod{p}$.

**Example 16.** Does the equation $x^2 + 2xy - 3y^2 = 0$ admit non-trivial rational solutions? The answer, yes! Though it is quite obvious taking $(1, 1) \in \mathbb{Q}^2$ naturally implies solutions in $\mathbb{Q}_p^2$ and $\mathbb{R}^2$. For sanity sake, let's check that Hasse-Minkowski also holds in the local fields. We already have a solution in $\mathbb{R}^2$, so it remains to check for $Q_p^2$ for all prime p. Consider $Q_2^2$, we have the equation modulo 2 as $x^2+y^2 \equiv 0 \pmod 2$, which has the non-trivial solution $(1, 1)$ in $Q_2^2$, $1+1 \equiv 0 \pmod 2$. Next consider $Q_3^2$ with the equation $x^2 + 2xy \equiv 0 \pmod 3$. Observe that $(1, 1) \in Q_3^2$ again is a solution. Finally, we check $Q_p^2$ for $p \geq 5$, where we now have the equation $x^2 + 2xy \equiv 3y^2 \pmod{p}$, which has again the non-trivial solution $(1, 1) \in Q_p^2$. Therefore, by Hasse-Minkowski Theorem, the equation does admit rational solutions, which we have already found at the start.

Perhaps a more illuminating example is when does it not admit a solution. Consider the example from our introduction.

**Example 17.** Does the equation $x^2 + y^2 - 3z^2 = 0$ admit non-trivial rational solutions? Let's check for solutions in $Q_3^3$. The equation is $x^2 + y^2 \equiv 0 \pmod 3$, and we can consider just the terms $0+0 \equiv 0 \pmod 3, 0+1 \not\equiv 0 \pmod 3, 1+1 \not\equiv 0 \pmod 3$ since any non-zero powers of 3 in the p-adic representation would be modded by mod 3. We see then the only solution for $x$ and $y$ is $x \equiv y \equiv 0 \pmod 3$. We can then write $x = 3a$ and $y = 3b$, for some $a, b \in Z_3$. Plugging this back into our original equation we have $(3a)^2 + (3b)^2 - 3z^2 = 0$, that is $9a^2 + 9b^2 = 3z^2$ or $z^2 = 3a^2 + 3b^2$. So taking mod 3, we have $z^2 \equiv 0 \pmod 3$, and the only solution is $z \equiv 0 \pmod 3$. Hence, we have only the trivial solution. Therefore, by Hasse-Minkowski Theorem, $x^2 + y^2 - 3z^2 = 0$ does not admit rational solutions.

## 5. Further Applications

Further applications of p-adic integers includes Monsky's Theorem on geometry, where he states that it is not possible to dissect a square into an odd number of triangles in equal area using 2-adic valuation [4, Ch. 22]. With what we saw of the natural embeddings in Section 3, the p-adic integers have also been regarded as a different way of completing the rational numbers, leading to study of p-adic analysis replacing $Q_p$ in place of $\mathbb{R}$ [2]. This has also led to the study of p-adic quantum mechanics, where p-adic forms an ultrametric space, leading to the study of discrete space-time rather than the usual continuous space-time. It has led to new solutions not possible in the standard case over $\mathbb{C}$ and even breaks the speed of light limit at which matter or energy may travel [5]. How fascinating.

## References

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